

Visual Anchor Point for Multimodal Document Retrieval

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Abstract

This report presents our solution to the **1st EReL@MIR Workshop Multimodal Document Retrieval Challenge Track** at *The Web Conference 2025*, where we achieved first place. The competition includes two retrieval tasks: *MMDocIR*, which focuses on retrieving relevant pages from long documents given multimodal queries, and *M2KR*, which aims to retrieve the most relevant document based on complex textual and visual inputs. To address these challenges, we propose a unified retrieval framework built upon the Generalist Multimodal Embedding (GME) model. Our approach incorporates several key components: (1) a $k \times [\text{EOS}]$ embedding strategy that improves representation quality by averaging multiple end-of-sequence token embeddings; (2) a visual-anchor-based adjustment mechanism that leverages visual similarity via DINO to identify highly relevant documents through matching illustrations; and (3) an ensemble learning strategy that combines multiple GME models trained on different data subsets and hyper-parameter configurations to enhance robustness and retrieval accuracy. Together, these components form a retrieval system that effectively handles multimodal and semantically complex queries. Our method demonstrates strong generalization across both tasks and outperforms existing baselines, ultimately earning us the top ranking in the challenge track.

1 Introduction

With the rapid development of Retrieval-Augmented Generation (RAG) techniques [15], multimodal documents have attracted growing attention in the field of information access and processing. Multimodal representation learning [7, 14], as a core supporting technology, has become increasingly prominent in recent years, especially in real-world applications such as e-commerce recommendation systems [16] (e.g., Amazon, eBay, Alibaba) and multimodal search engines [1, 8, 14] (e.g., Google, Bing, Baidu), where it demonstrates significant practical value.

The rapid progress in this research area has been largely driven by the success of large-scale Multimodal Foundation Models (MFMs), such as LLaMA [11] and GPT [13]. These models have set new benchmarks in tasks like web search, cross-modal retrieval, and

recommendation systems, pushing the boundaries of Multimodal Information Retrieval (MIR) and substantially improving system performance and generalization.

Against this backdrop, an increasing number of studies are exploring how to effectively integrate heterogeneous multimodal data sources—such as text and images [9] support downstream tasks in both generation and retrieval. The organization, modeling, and retrieval mechanisms for multimodal documents are becoming critical components of next-generation intelligent systems, particularly those powered by large foundation models.

To further explore the boundaries of multimodal document retrieval, we participated in the **EReL@MIR Workshop Multimodal Document Retrieval Challenge Track** [5] at *The Web Conference 2025*. This challenge aims to foster innovation in three key areas: Developing retrieval systems capable of processing diverse multimodal inputs; Designing efficient representation learning strategies tailored for multimodal content; Bridging academic research and industrial applications in multimodal retrieval and recommendation.

The competition consists of two tasks: **MMDocIR** [3] and **M2KR** [10]. The goal of MMDocIR is to retrieve the most relevant page(s) from a long document given a query consisting of text descriptions, images, or both. In contrast, M2KR focuses on retrieving the most relevant long document as a whole, based on a multimodal query.

Recent work has explored multimodal document retrieval using large language models (LLMs) [6, 18], which are increasingly capable of handling cross-modal signals. However, several challenges remain:

- **Knowledge Limitations:** Effective multimodal document retrieval requires web-level or domain-specific knowledge. While LLMs improve general understanding, they often struggle in specialized contexts, especially when visual inputs demand specific factual grounding. For example, if an image query shows a photo of Abraham Lincoln and the model fails to recognize it as such, it is unlikely to retrieve relevant documents about the U.S. Civil War.
- **Retrieval Biases:** A single retrieval model may introduce modal or semantic biases, making it hard to achieve consistently high retrieval precision across diverse queries.

To address these challenges, particularly the first, we propose a simple but effective idea inspired by human behavior: when users cannot directly understand an image, they often search for similar images within documents to determine relevance. We term this

approach the **Visual Anchor Method**, where related images in documents act as visual anchors to guide retrieval.

In this competition, we designed a retrieval framework incorporating the visual anchor strategy to compensate for the model's limited knowledge. We further employ a multimodal large language model (MLLM) [18] as the primary retriever to enhance world knowledge, and adopt ensemble learning techniques to improve robustness and retrieval accuracy.

2 Benchmarks

2.1 MMDocIR Benchmark

MMDocIR is a benchmark specifically curated for evaluating multimodal retrieval in long documents. It introduces two distinct tasks: page-level retrieval, focusing on identifying the most relevant pages within a document, and layout-level retrieval, targeting finer-grained elements such as paragraphs, tables, and figures. The dataset comprises 313 lengthy documents spanning ten domains, with annotations for 1,685 questions and over 170,000 bootstrapped labels [3]. MMDocIR emphasizes the challenges of integrating visual and textual information in complex document structures, highlighting the necessity for models that can effectively navigate and retrieve multimodal content.

2.2 M2KR Benchmark

The **Multi-task Multi-modal Knowledge Retrieval (M2KR)** benchmark is designed to evaluate general-purpose vision-language retrievers across diverse retrieval tasks. It encompasses three primary retrieval scenarios: Image-to-Text (I2T), Question-to-Text (Q2T), and Image&Question-to-Text (IQ2T). The benchmark integrates nine datasets, each tailored to assess specific aspects of multimodal retrieval capabilities [10]. M2KR serves as a foundational resource for training and evaluating models like PreFLMR, aiming to bridge the gap between visual and textual modalities in knowledge-intensive tasks.

3 Related Work

3.1 Multimodal Document Retrieval

Recent work in multi-modal document retrieval has demonstrated the effectiveness of vision-centric approaches that directly process document images. The *ColPali* model [4] introduces a novel retrieval paradigm by leveraging Vision-Language Models (VLMs) to encode document screenshots through a late interaction mechanism, bypassing traditional OCR and layout analysis pipelines. This approach maintains superior performance by preserving the original visual structure and relationships within documents. Similarly, [2, 3] shows that visual retrievers consistently outperform text-based methods, particularly for complex document elements like tables and figures where textual representations often fail to capture essential visual information. Both studies highlight the importance of maintaining raw visual features throughout the retrieval process, as intermediate text conversions inevitably lose critical document semantics. These findings suggest that future document retrieval systems should prioritize direct visual processing over conventional text-centric approaches.

3.2 Large Models for Multimodal Retrieval

Early approaches to multimodal retrieval predominantly utilized contrastive pretraining frameworks, such as CLIP [14], which align visual and textual modalities through dual-encoder architectures. These models, while effective in learning joint representations, often lack the comprehensive world knowledge and reasoning capabilities inherent in large-scale language models. Moreover, their architecture typically does not support the integration of complex multimodal inputs, limiting their applicability in scenarios requiring nuanced understanding across modalities.

Recent advancements have seen the emergence of Multimodal Large Language Models (MLLMs), which aim to bridge this gap by incorporating extensive pretraining on diverse datasets encompassing both visual and textual information. However, MLLMs are primarily optimized for generative tasks and may not inherently produce discriminative embeddings suitable for retrieval purposes.

To address this limitation, researchers have explored methods to adapt MLLMs for retrieval tasks. One such approach involves fine-tuning MLLMs using contrastive learning objectives on large-scale multimodal retrieval datasets, thereby enhancing their ability to generate discriminative embeddings from their last hidden states. For instance, the GME (Generalist Multimodal Embedding) model [18] is built upon the Qwen2-VL [17] architecture and is trained to produce unified embeddings for text, images, and image-text pairs. This adaptation enables the model to perform effectively in retrieval tasks across different modalities.

In our work, we adopt the GME model as our retriever, leveraging its capability to handle diverse multimodal inputs and generate robust embeddings suitable for retrieval tasks. This choice is motivated by GME's demonstrated performance in bridging the gap between generative pretraining and discriminative retrieval objectives.

4 Method

4.1 Total Framework

Overall, we design an ensemble-based retrieval framework. We fine-tune five GME [18] models with different parameter configurations. Each model produces an independent ranking list R_i for a given query. We aggregate the final ranking by averaging the ranks across all models.

In addition to this ensemble strategy, we introduce a visual-anchor-based adjustment. For each image in the query, we compute its visual similarity with all candidate visual anchors extracted from the documents using DINO [12]. If the most similar anchor has a similarity score exceeding a predefined threshold θ , we directly promote the corresponding document (containing the anchor) to the top of the final ranking list.

This hybrid approach enables the system to leverage both the strengths of MLLM-based representation and the robustness of vision-level anchor matching. For the MMDocIR task, we follow the baseline approach proposed by [?], which searches over document screenshots using text-based queries. For M2KR we employ visual anchor point strategy.

4.2 Visual Anchor Point

In our setup, we treat all images embedded within documents as *visual anchors*.

Considering real-world scenarios, queries are often rich in content and involve highly domain-specific information. For example, a query image showing Abraham Lincoln requires the model not only to understand what the person looks like, but to recognize that it is indeed Lincoln and retrieve documents related to his identity (e.g., the American Civil War). Retrieval models that rely purely on encoder-based semantic representations often fail in such cases, as these encoders tend to abstract visual features into general concepts (e.g., “a man in formal attire”) rather than preserving fine-grained, instance-level identity.

To address this issue and avoid semantic loss from visual abstraction, we draw inspiration from how humans identify relevant documents—often through recognizing similarities between the query image and illustrations within documents. We formalize this observation by introducing the **visual anchor** method.

Specifically, we use DINO [12] to compute the similarity between the query image and all document visual anchors. If the most similar anchor exceeds a similarity threshold θ , we consider the corresponding document to be the correct (ground truth) target for retrieval. This method enables direct matching at the visual level, effectively circumventing limitations of abstracted semantic embeddings in encoder-only models.

4.3 Improved Representation via Multiple [EOS] Tokens

Most existing MLLM-based retrieval systems extract the document or query embedding by appending a single [EOS] token to the input and using its final hidden state as the representation [6]. However, relying on a single token can be limiting—especially for complex multimodal queries that contain intricate visual-textual relationships or fine-grained semantic nuances. A single token may fail to fully capture the diversity of information encoded across different parts of the input.

To address this, we adopt a simple yet effective strategy: instead of appending a single [EOS] token, we append $k \in [5, 10]$ [EOS] tokens to the input. At inference time, we average the embeddings of these k tokens to form the final representation. This approach can be seen as performing a soft aggregation over the tail end of the transformer sequence, which is known to contain distilled high-level features.

In empirical evaluations, this $k \times [\text{EOS}]$ embedding strategy consistently improved retrieval performance. Specifically, it increased top- k recall by 2–3 percentage points, without adding any significant computational overhead during inference.

4.4 Ensemble Learning

To curb single-model bias and improve robustness, we train *five* GME retrievers, each on a different sample of the training data of M2KR and MMdocIR and with slightly varied hyper-parameters (learning rate, epochs, batch size, dropout). At inference, every model outputs a ranked list; we convert ranks to reciprocal scores and average them, yielding a fused ranking that consistently boosts recall without extra computational cost.

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